



1.8

Rational Functions and Zeros

Student Notes and Guided Practice

AP Precalculus - Unit 1

| ECHS Mathematics

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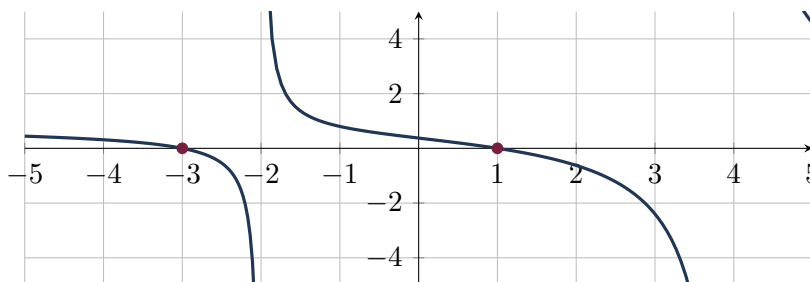
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Learning Targets

- Determine rational-function zeros and distinguish them from excluded values.
- Use factored form to connect intercepts and sign behavior.
- Track canceled factors and domain restrictions.

Essential Question

How do numerator factors, denominator factors, and cancellations interact?



Concept Framework

Key Idea

A zero occurs where the simplified numerator is zero and the original denominator is nonzero.

Key Idea

Denominator zeros are excluded from the domain.

Key Idea

A common factor may create a hole rather than a zero or vertical asymptote.

Key Idea

Sign charts use all critical numbers: zeros and excluded inputs.

Formula Map

Formula and Structure

$$R(x) = 0 \iff P(x) = 0 \text{ and } Q(x) \neq 0$$

Formula and Structure

$$\text{Dom}(R) = \{x : Q(x) \neq 0\}$$

Worked Examples

Worked Example 1

Problem. Find the zeros of $R(x) = \frac{(x-4)(x+1)}{(x+1)(x-2)}$.

Solution. The original domain excludes -1 and 2 . After cancellation, only $x = 4$ is a zero; $x = -1$ is a hole.

Worked Example 2

Problem. Determine the sign of $\frac{x-3}{(x+2)(x-1)}$ on $(1, 3)$.

Solution. Test $x = 2$: numerator negative; both denominator factors positive; the quotient is negative.

Worked Example 3

Problem. Can an excluded input also be listed as an x-intercept?

Solution. No. An x-intercept must be a point on the graph, while an excluded input is not in the domain.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. For $\frac{(x-3)(x+2)}{(x-1)(x+2)}$, find zeros and excluded inputs and classify any cancellation.

2. For $\frac{x^2-9}{x^2-4}$, find zeros and excluded inputs and classify any cancellation.

3. For $\frac{(2x+1)^2}{(x-4)^3}$, find zeros and excluded inputs and classify any cancellation.

4. For $\frac{x(x-5)}{x^2+1}$, find zeros and excluded inputs and classify any cancellation.

5. For $\frac{(x+1)(x-2)}{(x+1)^2}$, find zeros and excluded inputs and classify any cancellation.

6. Make a sign chart for $R(x) = \frac{(x-2)(x+1)}{(x-3)(x+4)}$.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.